Assignment 5: Materials and Methods (Methodology Section)

Course: NLP with Deep Neural Networks
Type: Individual Submission
Submission: Overleaf + Canvas (PDF)

Overview

This assignment builds directly on your Abstract and Introduction. Your goal is to write a clear, detailed, and reproducible Materials and Methods section for your project. This section should answer one key question: "How exactly will your system be built, trained, and evaluated?". A strong Methods section should allow another researcher to replicate your work without guessing.

Submission Instructions

You are required to:

1. Update your Overleaf document
2. Add and complete a new section titled: **"Materials and Methods"**
3. Upload the updated PDF to Canvas

Section Requirements

Your Materials and Methods section should be **2–3 pages** and must include the following:

1. Dataset and Data Processing

Clearly describe your data.

Include:

- Dataset name(s) and source(s)
- Type of data (text, structured, multimodal, etc.)
- Size (number of samples, classes, etc.)
- Data collection method (API, scraping, public dataset, etc.)
- Preprocessing steps:
    - Cleaning (removal, normalization, filtering)
    - Tokenization / embedding preparation

    - Handling noise, missing values, or imbalance

Important note: Be specific. Avoid vague statements like "data is cleaned."

2. Problem Formulation

Define your task precisely.

Include for example:

- Task type (classification, generation, retrieval, etc.)
- Input → Output format
- Labels (if supervised)
- Any assumptions or constraints

3. Model / System Architecture

Describe your approach.

Depending on your project, include for example:

- Model(s) used:
    - Classical (TF-IDF, FFNN)
    - Deep Learning (LSTM, Transformer, etc.)
    - LLM-based (prompting, fine-tuning, RAG, agents)
- Architecture details:
    - Key components
    - Pipeline stages
- Tools/frameworks:
    - PyTorch, TensorFlow, HuggingFace, LangChain, etc.

Important: Explain *why* this approach is suitable (briefly, not theoretical).

4. Training Strategy

Explain how your model learns.

Include, for example:

- Train/validation/test split
- Training procedure:
    - Epochs, batch size, optimizer (if applicable)
- Fine-tuning / prompting strategy (if using LLMs)
- Any hyperparameters (only key ones)

5. Evaluation Framework

Explain how you will measure performance.

Include, for example:

- Metrics:
    - Classification → Accuracy, Precision, Recall, F1
    - Generation → BLEU, ROUGE, human evaluation
    - Retrieval → Recall@k, MRR
- Baselines:
    - What are you comparing against?
- Evaluation setup:
    - Test cases, validation strategy
- (Optional but encouraged) Error analysis plan

6. Implementation Details (Brief)

Include:

- Libraries/tools used
- Hardware (if relevant: GPU/CPU)
- Any constraints or assumptions

Mathematical Formulation (Optional but highly Encouraged)

You may include mathematical expressions if they help clarify your method or model.

Where to include:

- Primarily in Model / System Architecture
- Optionally in:
    - Training Strategy (e.g., loss function)
    - Evaluation Framework (only if needed)

What to include (if applicable):

- Loss functions (e.g., cross-entropy)
- Model computations (e.g., attention, scoring functions)
- Similarity measures (e.g., cosine similarity)

Guidelines:

- Keep it minimal and directly tied to your implementation
- Do NOT include long derivations or theory
- Clearly define symbols used
- Briefly explain what the equation represents

When it is NOT required:

- Purely system-based projects (e.g., RAG pipelines, agents, API workflows)
- Projects focused on implementation rather than model design

Important Note: Mathematical formulation is not required for full credit, but when used appropriately, it strengthens technical clarity and presentation.

Writing Guidelines

- Be **clear, structured, and technical**
- Avoid unnecessary theory
- You can use AI for polishing purposes after you have written the initial draft

Rubric Overview

Total: 100 Points

1. Dataset & Preprocessing (15 Points)

- Clear dataset description (5)
- Preprocessing steps well-defined (5)
- Data handling is realistic and appropriate (5)

2. Problem Formulation (10 Points)

- Task clearly defined (4)
- Input-output mapping clear (3)
- Labels/constraints specified (3)

3. Model / System Design (25 Points)

- Model(s) clearly described (10)
- Architecture/pipeline well-structured (10)
- Approach is logical and appropriate (5)

4. Training Strategy (15 Points)

- Training process clearly explained (5)
- Key hyperparameters mentioned (5)
- Strategy is feasible (5)

5. Evaluation Framework (20 Points)

- Metrics are appropriate (8)
- Baselines/comparisons included (6)
- Evaluation setup clearly defined (6)

6. Implementation Details (5 Points)

- Tools/frameworks clearly mentioned (3)
- Practical setup described (2)

7. Writing Quality & Clarity (10 Points)

- Clear, structured, professional writing (10)

Important Notes

- This is a **technical section**, not a literature review
- Focus on **clarity and reproducibility**, not storytelling
- Avoid vague phrases like:
    - “we use AI model”
    - “we preprocess the data”
- Be precise enough that someone could **implement your system from your description**